

.....
.....
.....
3) What is a swap?
.....
.....
.....

21.4 MODELS OF DERIVATIVE PRICING

Derivative pricing is a complex process as it involves determining the fair value of a derivative contract. The pricing of derivatives depends on various factors, including the type of derivative, the characteristics of the underlying asset, market conditions, and the methodologies used. This section discusses two popular derivative pricing models: **Binomial Option Pricing Model** and **Black-Scholes Option Pricing Model**.

21.4.1 Binomial Option Pricing Model

The binomial options pricing model, initially introduced by Cox, Ross, and Rubinstein in 1979, offers a versatile numerical approach to valuing options. This model employs a “discrete-time” framework to track the evolving price of the underlying financial instrument over time, facilitating option valuation through the application of the risk neutrality assumption.

Unlike other models, the binomial options pricing model excels in handling diverse conditions, including American options that can be exercised at any point and Bermudan options exercisable at various points. Its simplicity and mathematical feasibility make it easily implementable in software or spreadsheet environments. While it may be slower than the Black-Scholes model, practitioners often prefer it for its accuracy, especially in valuing longer-dated options and those associated with dividend payments.

For options involving multiple uncertainties (e.g., real options) or complex features (e.g., Asian options), lattice methods encounter challenges, making Monte Carlo option models more suitable. However, Monte Carlo simulation, although comprehensive, is computationally intensive and not the preferred choice when lattice methods or formulas are adequate.

The binomial pricing model operates within a “discrete-time framework,” utilizing a binomial lattice (tree) to trace the evolution of the option’s underlying variable across multiple time steps. Each node in the lattice represents a potential price of the underlying at a specific point in time, forming the basis for option valuation. The valuation process involves three key steps:

- 1) Price tree generation: Creating a tree to depict the evolution of the underlying asset's price.
- 2) Calculation of option value at final nodes: Determining the option's intrinsic (exercise) value at the expiration of the option.
- 3) Progressive calculation of option value at earlier nodes: Iteratively calculating the option value at each preceding node using the risk neutrality assumption, where the fair price today is equal to the discounted expected value of its future payoff. The "Binomial Value" at each node is determined by weighting the option values from the later two nodes by their respective probabilities of an up or down move and discounting them at the risk-free rate.

For an American option, which can be held or exercised before expiry, the value at each node is the maximum of the Binomial Value and the Exercise Value. The Binomial Value is computed using the formula:

$$\text{Binomial Value} = [p \times \text{Option up} + (1-p) \times \text{Option down}] \times \exp(-r \times t)$$

Here, p represents the probability of an up move in the underlying,

$(1-p)$ is the probability of a down move,

r is the risk-free rate,

and t is the time to expiration.

This process is repeated for each node, concluding at the valuation date, where the final result represents the option's value.

21.4.2 The Black-Scholes Option Pricing Model

The Black-Scholes model serves as a mathematical framework for evaluating the evolving prices of financial instruments, particularly stocks. Fischer Black and Myron Scholes developed the Black-Scholes formula in 1973, building upon earlier work by Edward Thorpe, Paul Samuelson, and Robert C. Merton. This model is widely used in practice to value call and put options. The model is based on the following assumptions:

- 1) The underlying share/stock does not generate dividends during the life of the call option.
- 2) Transaction costs are zero.
- 3) The risk-free rate of return is known and it remains constant during the life of the option.
- 4) The option can be exercised at the expiry only not before that. It means the option is a European not American.
- 5) Short selling is allowed and the short seller will get immediately the full cash proceeds of today's price for a security sold short.
- 6) Stock price moves randomly.

If the above assumptions are met, then as per the Black-Scholes Option Pricing Model, the value of a call option can be calculated using the following formula:

$$VCO = P[N(d_1)] - Ye^{-rt}[N(d_2)]$$

Such that

$$d_1 = \frac{\ln(P/Y) + [r + (\sigma^2/2)]t}{\sigma\sqrt{t}}$$

$$d_2 = d_1 - \sigma\sqrt{t}$$

where,

VCO = Value of a call option

P = Current price of the underlying share or stock

Y = Exercise/strike price of the option

r = Risk – free rate of return (should be continuously compounded)

t = Time to the expiry of the call option (should be an annual figure and used in fraction)

σ^2 = Annualised variance of the rate of return on the underlying stock or share

$N(d_1)$ and $N(d_2)$ represent the cumulative areas under a standard normal distribution i.e., a normal distribution with zero mean and standard deviation 1.

$\ln(P/Y)$ = Natural log of (P/Y)

$e = 2.7183$ (Approximately)

If we want to calculate the value of a put option, then the formula is

$$VPO = Ye^{-rt}[1 - N(d_2)] - P[1 - N(d_1)]$$

Example 21.1: Suppose we want to value a call option where the underlying asset is the share of Tata Consultancy Service. The exercise price of the option is ₹ 20 and the current price of the share is ₹ 25. Further, suppose that the treasury bills rate (a proxy used for risk-free rate) is 5%, the annualised variance of the returns on the share is 0.16, and the call option will expire at the end of 3 months. The value of the call option can be calculated as follows:

Solution

Step 1: Summarise the information

$P = ₹ 25$

$$Y = ₹ 20$$

$$r = 5\% \text{ or } 0.05$$

$$t = 3 \text{ months or } 3/12 = 0.25 \text{ years}$$

$$\sigma^2 = 0.16 \Rightarrow \sigma = 0.4$$

Step 2: Find out d_1 and d_2

$$d_1 = \frac{\ln(P/Y) + [r + (\sigma^2/2)]t}{\sigma\sqrt{t}}$$

$$d_1 = \frac{\ln\left(\frac{25}{20}\right) + \left[0.05 + \left(\frac{0.16}{2}\right)\right] 0.25}{0.4\sqrt{0.25}}$$

$$= \frac{\ln(1.25) + (0.13) \times 0.25}{0.4 \times 0.5}$$

Using Excel/Scientific calculator/log table, we can calculate the natural log of 1.25 i.e., $\ln(1.25) = 0.2231$

$$d_1 = \frac{0.2231 + 0.0325}{0.2} = \frac{0.2556}{0.2}$$

$$d_1 = 1.278$$

Therefore,

$$d_2 = d_1 - \sigma\sqrt{t} = 1.278 - 0.4\sqrt{0.25} = 1.278 - 0.2 = 1.078$$

Step 3: Find $N(d_1)$ and $N(d_2)$

Referring to the Standard Normal Distribution Table, we find the cumulative probabilities i.e.,

$$N(d_1) = N(1.278) = 0.8994$$

$$N(d_2) = N(1.078) = 0.8595$$

These cumulative probabilities can also be calculated using MS Excel. You should use the following Excel formula “= NORM.DIST(d_1 ,0,1, True)”.

Now, the value of the call option as per the Black-Scholes Option Pricing Model is

$$VCO = P[N(d_1)] - Ye^{-rt}[N(d_2)]$$

$$VCO = 25(0.8994) - 20 \times e^{-0.05 \times 0.25}(0.8595)$$

$$= 22.485 - 20 \times e^{-0.0125}(0.8595)$$

We can calculate the value of $e^{-0.0125}$ using the log technique or scientific calculators or in MS-Excel using the formula “= POWER (2.7183, -0.0125)”. We find that $e^{-0.0125} = 0.9875$. Thus,

$$VC0 = 22.485 - 20 \times 0.9875 \times (0.8595)$$

$$= 22.485 - 16.97513$$

$$VCO = 5.509875 \approx 5.5099$$

Thus, the call option value is ₹ 5.5099.

Suppose it was a **put option**, then

$$VPO = Ye^{-rt}[1 - N(d_2)] - P[1 - N(d_1)]$$

$$VPO = 20 \times 0.9875 \times [1 - 0.8595] - 25[1 - 0.8994]$$

$$= 2.7748 - 2.515$$

$$= 0.2598$$

Advantages & Disadvantages of the Black-Scholes Model:

Advantage: One major advantage of the Black-Scholes Model is its speed, allowing for the rapid calculation of a large number of option prices.

Limitation: The model's significant limitation is its inability to accurately price American-style options, as it only calculates prices at expiration and does not account for potential early exercise steps.

Assumptions underlying the Black-Scholes model include the absence of arbitrage opportunities, the possibility of short selling at all times, no transaction costs or taxes, perfect divisibility of securities, continuous trading, and no dividends paid by the underlying stock during the option's lifetime. Additionally, it assumes that the risk-free rate (r) and stock volatility (σ) are known over the option's lifetime.

21.4.3 Relation between the Binomial Options Pricing Model & Black-Scholes Model

Both the binomial and Black-Scholes models share a foundation rooted in the assumption that stock prices follow a stochastic process described by geometric Brownian motion. For European options, the binomial model approaches the Black-Scholes formula as the number of binomial calculation steps increases. In essence, the Black-Scholes model for European options can be seen as a special case of the binomial model with an infinite number of steps, representing a continuous process approximation. Therefore, the binomial model offers discrete approximations to the continuous process underlying the Black-Scholes model.

- 1) How will you determine the value of call and put options using the Black- Scholes Option Pricing Model?

.....

- 2) The current price is ₹ 35, the exercise price is ₹ 35, the risk-free rate of interest is 10%, the volatility of returns of the underlying asset is 20% and the time to expiry is 1 year. Calculate the value of the call option using the Black-Scholes Option Model. What would be your answer if it were a put option, not a call option? (**Hint:** Refer to example 4).

.....

Note: Students can use an online calculator (scan the QR code) to value options as per the Black-Scholes Option Pricing Model.



21.5 MARKET OF DERIVATIVES IN INDIA

India has two types of derivative markets: the exchange-traded market and the over-the-counter (OTC) market. In the exchange-traded market, standardized contracts are traded on an exchange. In contrast, the OTC market is decentralized, with contracts negotiated directly between two parties. SEBI is the regulator of the derivative market.

Trade in equity and index derivatives

If you want to trade in equity and index derivatives, you can do so through the National Stock Exchange (NSE) or Bombay Stock Exchange (BSE). The derivative market in India started in the early 1990s. The first derivatives contract was launched on the Bombay Stock Exchange (BSE) in June 2000. Since then, the derivative market in India has grown exponentially¹. There

¹ <https://www.bsebti.com/blog/derivative-market-in-india-meaning-types-participants-differences/#:~:text=Difference%20Between%20Cash%20and%20Derivative%20Market&text=The%20derivative%20market%20in%20India,in%20India%20has%20grown%20exponentially.>

are two types of markets – the cash market and the derivative market. The difference between cash and derivative market is that transactions are settled on the spot in the cash market. In contrast, in the derivative market, transactions are settled at a future date. The participants in the cash market are known as spot traders, while the participants in derivative market are known as futures traders. Spot trading is done in the physical form of commodities, whereas futures trading is done via derivatives contracts.

The National Stock Exchange of India Limited (NSE) commenced trading in derivatives with the launch of index futures on June 12, 2000². The futures contracts are based on the popular benchmark Nifty 50 Index. The Exchange introduced trading in Index Options (also based on Nifty 50) on June 4, 2001. NSE also became the first exchange to launch trading in options on individual securities from July 2, 2001. Futures on individual securities were introduced on November 9, 2001. Futures and Options on individual securities are available on securities that meet the eligibility criteria stipulated by SEBI. The Exchange has also introduced trading in Futures and Options contracts based on Indices. Currently, Derivatives on NIFTY 50, NIFTY Bank, NIFTY Financial Service, NIFTY Midcap Select and NIFTY Next 50 are available for trading. Since the launch of the Index Derivatives on the popular benchmark Nifty 50 Index in 2000, the National Stock Exchange of India Limited (NSE) today have moved ahead with a varied product offering in equity derivatives. The Exchange currently provides trading in Futures and Options contracts on – 5 major indices and more than 180 securities.

In India, both the BSE and NSE provide derivatives trading facilities. The NSE's derivatives platform is known as Nifty Futures and Options (F&O). The BSE offers F&O trading on its own exchange and also facilitates trading on the NSE's exchange.

Trading in Commodity Derivatives

If you want to trade in commodities like crude oil, gold, metals, etc., you can do so through commodity exchanges like the Multi Commodity Exchange (MCX) or the National Commodity and Derivatives Exchange (NCDEX) both under the ownership of Ministry of Finance, Government of India. Conversely, if you want to trade in currencies, Metropolitan Stock Exchange of India (MSE) Limited under the ownership of Ministry of Finance, Government of India, simplifies that.

21.6 LET US SUM UP

This unit builds upon the topics explored in units 13, 18 and 19, focusing on assets and their pricing. The discussion centres around a specific type of asset whose value is intricately tied to and influenced by the price movements of

² <https://www.nseindia.com/products-services/about-equity-derivatives>

an underlying asset. The unit commences with an introduction to the fundamental concept and rationale behind derivatives. Subsequently, it delves into an extensive examination of various types of derivatives, encompassing forward contracts, futures, and swaps.

The unit then shifts its focus to the pricing mechanisms of derivatives, providing insights into both the binomial model and the Black-Scholes formula. A comparative analysis is conducted to highlight the distinctions between these two models.

21.7 KEY WORDS

Arbitrageurs	: Investors engaging in arbitrage to exploit price discrepancies in securities to earn riskless returns.
Call Option	: A financial contract providing the holder the right to purchase an asset at a predetermined price within a specified timeframe.
Forward Contract	: A mutual agreement between two parties to exchange an asset for cash at a predetermined future date, with the current price fixed at the time of agreement.
Hedge	: An investment made to offset potential losses in another investment, creating a balance between the two.
Interest Rate Future	: Futures contracts based on fixed-income securities like treasury bills, bonds, and CDs.
Interest Rate Options	: Options contracts related to fixed-income securities such as treasury bills.
Interest Rate Parity	: The concept suggesting that the difference in interest rates between two countries equals the disparity between the forward and spot rates of those respective countries.
Long Position	: The position of an investor entitled to receive an asset in the future, representing the buyer's role in the security.
Margin	: A portion of the transaction value that the customer must pay to initiate a transaction, with the remaining amount often borrowed from the broker. Two types of margins exist: initial margin and maintenance margin.

Option Contract	:	A contractual agreement providing the right to buy or sell an asset at a predetermined price on a future date, with the option buyer having the discretion to exercise this right.
Put Option	:	An option granting the holder the right to sell an asset at a fixed price within a specified period.
Short Position	:	The position of an investor obligated to deliver an asset in the future, indicating the seller's role in the security.
Spot Rate	:	The exchange rate applicable to the immediate delivery of currency, typically settling two days after the trade date.
Swap Contract	:	An agreement between two parties (counterparties) to exchange assets or a series of cash flows over a specified period at predetermined intervals.

21.8 SOME USEFUL BOOKS

Bhole, L M . (2004) *Financial Institutions and Markets 4th edition*, TataMcGraw- Hill, New Delhi:

Dothan, U. (1990), *Prices in Financial Markets*, Oxford University Press, New York

Hull, John, (2002) *Options, Futures and other Derivatives* Prentice-Hall of India, New Delhi.

“Options, Futures, and Other Derivatives” by John C. Hull

21.9 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Derivatives encompass contracts such as forwards, futures, options, or hybrids, with predetermined durations, linked to the value of specified real or financial assets or an index of securities.
- 2) Risk transfer from risk-averse to risk-oriented individuals, price discovery for future and current prices, stimulation of entrepreneurial activity, increased market trading volumes due to the participation of risk-averse individuals, and promotion of savings and long-term investments.

Check Your Progress 2

- 1) Agreement between two entities, culminating in settlement on a predetermined future date at an agreed-upon current price.
- 2) Futures Contracts are typically traded on exchanges. In a futures transaction, two parties negotiate through brokerage firms that hold a “seat” on the exchange for that particular contract.
- 3) A swap is a contractual agreement between two parties to exchange assets, financial obligations, or cash flows over a specified period at predetermined intervals.

Check Your Progress 3

- 1) Refer to Sub-section 21.4.2
- 2) The current price is ₹ 35, the exercise price is ₹ 35, the risk-free rate of interest is 10%, the volatility of returns of the underlying asset is 20% and the time to expiry is 1 year. Calculate the value of the call option using the Black-Scholes Option Model. What will be your answer if it were a put option not call option?

Step 1: Summarise the information

$$P = ₹ 35$$

$$Y = ₹ 35$$

$$r = 10\% \text{ or } 0.10$$

$$t = 1 \text{ year}$$

$$\sigma^2 = 0.04 \Rightarrow \sigma = 0.20$$

Step 2: Find out d_1 and d_2

$$d_1 = \frac{\ln(P/Y) + [r + (\sigma^2/2)]t}{\sigma\sqrt{t}}$$

$$d_1 = \frac{\ln\left(\frac{35}{35}\right) + \left[0.10 + \left(\frac{0.04}{2}\right)\right] 1}{0.20\sqrt{1}}$$

$$= \frac{\ln(1) + (0.12)}{0.20}$$

Using Excel/Scientific calculator/log table, we can calculate natural log of 1
i.e., $\ln(1) = 0$

$$d_1 = \frac{0 + 0.12}{0.20} = \frac{0.12}{0.20}$$

$$d_1 = 0.6$$

Therefore,

$$d_2 = d_1 - \sigma\sqrt{t} = 0.6 - 0.20\sqrt{1} = 0.6 - 0.20 = 0.4$$

Step 3: Find $N(d_1)$ and $N(d_2)$

Referring the Standard Normal Distribution Table, we find the cumulative probabilities i.e.,

$$N(d_1) = N(0.6) = 0.7257$$

$$N(d_2) = N(0.4) = 0.6554$$

These cumulative probabilities can also be calculated using MS-Excel. You should use the following excel formula “= NORM.DIST ($d_1, 0, 1$, True)”.

Now, the value of the call option as per the Black-Scholes Option Pricing Model is

$$VCO = P[N(d_1)] - Ye^{-rt}[N(d_2)]$$

$$VCO = 35(0.7257) - 35 \times e^{-0.10}(0.6554)$$

$$= 35 \times 0.7257 - 35 \times 0.9048 \times 0.6554 = 4.644$$

Note: We have calculated the value of $e^{-0.10}$ using log technique or scientific calculators or in MS-Excel using the formula “= POWER (2.7183, -0.10)”. We find that $e^{-0.10} = 0.9048$.

Thus, the call option value is ₹ 4.644.

Suppose that it were a **put option**, then

$$VPO = Ye^{-rt}[1 - N(d_2)] - P[1 - N(d_1)]$$

$$VPO = 35 \times 0.9048 \times [1 - 0.6554] - 35[1 - 0.7257]$$

$$= 11.91024 - 9.6005$$

$$= 1.312$$